

# Recursive Offline Distillation via Epistemic Complexity Approximation

Anonymous ACL submission

## Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate epistemic uncertainty using a medium model (70-72B) as an aleatoric proxy and sample only a small subset of the reasoning chains for distillation. Our approach consistently outperforms naive distillation on comparable computational budget ( $57.15 \pm 1.01$  vs 51.61 on average) and offers a much steeper learning curve. Additionally, we present a comparative analysis of methods to generate and normalize distillation traces. We publish our code and data<sup>1</sup>.

## 1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them demonstrate exceptional results and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026) that can be fine-tuned and deployed in resource-constrained environments.

A prominent approach for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher (Agarwal et al., 2024; Jiao et al., 2020; Gu et al., 2024), which is not always possible (Zhou

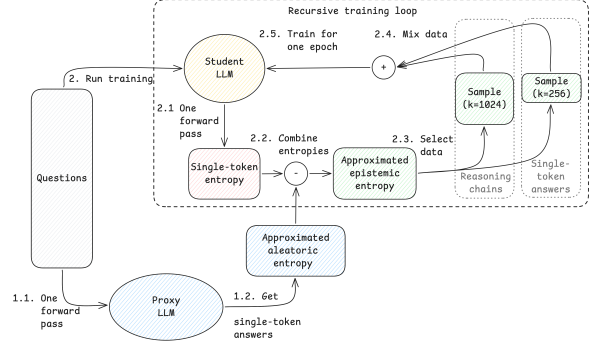


Figure 1: Recursive offline distillation scheme

et al., 2023; Ryskulov et al., 2026). It forces practitioners to explore offline distillation, where the student model is trained on the pre-generated teacher’s text output (Li et al., 2023; Hsieh et al., 2023).

We propose a fully automated recursive pipeline for offline distillation (Figure 1, described in 3) based on uncertainty decomposition (Wang et al., 2025b). The method consistently outperforms naive distillation ( $57.15 \pm 1.01$  vs 51.61 on average), and it is designed to work in resource-constrained environments.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

## 2 Related Work

Recent work improved distillation through training objectives (Ko et al., 2024; Xu et al., 2026), data selection (Just et al., 2026; Li et al., 2025; Yang et al., 2026), and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026). Joint supervision with rationales and labels improves data efficiency (Hsieh et al., 2023). Curriculum benefits vary with model capability, task complexity, and the choice

<sup>1</sup>TODO

of difficulty metric (Jia et al., 2026). ELAD (Zhang et al., 2024) selects uncertain student explanations for teacher revision. Complexity-aware fine-tuning (Goncharov et al., 2026) selects reasoning examples by single-token entropy. We recursively sample pre-generated traces using student-proxy entropy differences.

Trace synthesis includes explaining known solutions (Li and Mooney, 2024; Wang et al., 2025a) and learning from corrections (An et al., 2024; Yang et al., 2025). We compare direct generation and various augmentations on the same task under constrained trace lengths.

### 3 Method

**Our central hypothesis** is that we can approximate epistemic uncertainty of the student model with its entropy and entropy of a larger LLM. Goncharov et al. (2026) show that a single-token entropy (Figure 4) is a reasonable complexity predictor for a given entry in a question-answering dataset. We assume that it approximates the total uncertainty and that we can decompose it into epistemic and aleatoric components (Depeweg et al., 2018; Malinin and Gales, 2021). With scaling laws (Kaplan et al., 2020) in mind, we hypothesize that larger models should have lower total uncertainty. Therefore, in comparison to a small student model (3-4B) its substantially larger sibling (70-72B<sup>2</sup>) could be treated as a weak aleatoric proxy.

For a question  $x$ , assume  $H(x) = A(x) + E(x)$ , where  $H_S$ ,  $H_P$  - single-token entropy (student and proxy),  $A$  - shared aleatoric component, and  $E_S$ ,  $E_P$  - model-specific epistemic component. If proxy’s residual epistemic uncertainty satisfies  $E_P(x) \leq \varepsilon(x)$ , then

$$\begin{aligned} A(x) &= H_P(x) - E_P(x) \geq H_P(x) - \varepsilon(x) \\ E_S(x) &= H_S(x) - A(x) \leq H_S(x) - H_P(x) + \varepsilon(x) \end{aligned} \quad (1)$$

Thus, the entropy difference (*entropy gain*) approximates the student’s epistemic uncertainty with error at most  $\varepsilon(x)$ .

**The proposed pipeline** consists of the following steps (Figure 1, Algorithm 1): (1) prompt the larger model to answer all questions in the dataset with a single token once (or an ensemble of models taking average), collect entropy, assume it is an aleatoric uncertainty proxy; (2) for each training

#### Algorithm 1 Recursive offline distillation via epistemic complexity approximation

---

**Require:** Student  $S$ , proxy model  $P$ , epochs  $T$   
**Require:** Questions  $D$ , ground-truth answers  $y_x$ , and pre-computed reasoning traces  $r_x$   
**Ensure:** Trained student  $S$

```

1: for all  $x \in D$  do
2:    $H_P(x) \leftarrow \text{ENTROPY}(P, x)$ 
3: end for
4: for  $t = 1, \dots, T$  do
5:   for all  $x \in D$  do
6:      $H_S(x) \leftarrow \text{ENTROPY}(S, x)$ 
7:      $\text{EG}(x) \leftarrow \max\{H_S(x) - H_P(x), 0\}$ 
8:   end for
9:    $Q \leftarrow \text{WEIGHTEDSAMPLE}(D, \text{EG}, 1024)$ 
10:   $Q_A \leftarrow \text{first 256 questions in } Q$ 
11:   $B_R \leftarrow [(x, r_x) : x \in Q]$ 
12:   $B_A \leftarrow [(x, y_x) : x \in Q_A]$ 
13:   $S \leftarrow \text{TRAIN}(S, B_R \parallel B_A)$ 
14: end for
15: return  $S$ 

```

---

epoch, (2.1) prompt the student model to answer all questions in the dataset with a single token, collect entropy; (2.2) calculate entropy gain  $\text{EG}(x) = \max\{H_S(x) - H_P(x), 0\}$ ; (2.3.1) sample 1024 synthetic reasoning chains without replacement, weighted by positive entropy gain; (2.3.2) sample 256 single-token answers in similar fashion as a subset of the previous sample; (2.4) concatenate samples; (2.5) train the student using LoRA (Hu et al., 2021) on reasoning traces first, followed by single-token responses.

## 4 Results

### 4.1 Experimental Setup

**Dataset:** Multiple choice question answering dataset MMLU Pro (Wang et al., 2024). It has questions of different complexities from 14 domains, ten options for each, single correct one. Random train/test split - 80/20.

**LLMs:** Aleatoric proxy: ensemble of two medium models<sup>3</sup> (taking average of their entropies) - LLaMA 3.3 70B (Grattafiori et al., 2024), Qwen 2.5 72B (Qwen-Team, 2024). Students: three small models Qwen2.5 3B (Qwen-Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024). Reasoning trace source: DeepSeek V4 Pro and Flash (DeepSeek-AI et al., 2026) via API.

<sup>2</sup>This particular size is chosen based on the imposed resource constraints (80GB VRAM) and open model availability

<sup>3</sup>All models and datasets are published under permissive licenses that allow them to be used for research purposes

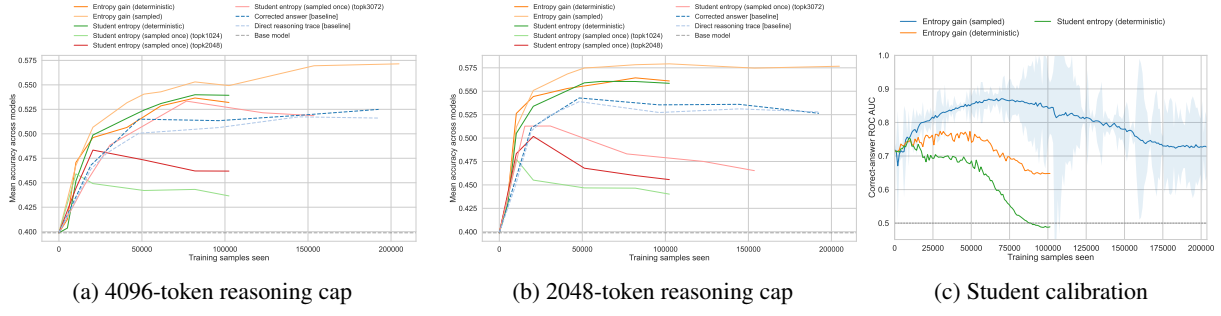


Figure 2: Distillation pipeline results across three models (average) on MMLU-Pro

| Method                                      | LLaMA-3B                                          | Qwen-3B                                           | Phi4-mini                                  | Mean                                              |
|---------------------------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------|---------------------------------------------------|
| Direct reasoning trace                      | 46.09 / 45.55                                     | 51.45 / 50.37                                     | 57.27 / 62.43                              | 51.61 / 52.78                                     |
| Corrected answer                            | 47.80 / 47.26                                     | 50.87 / 49.42                                     | <u>58.81</u> / 61.26                       | 52.49 / 52.65                                     |
| Student entropy (deterministic)             | 49.83 / 50.08                                     | 53.62 / 53.03                                     | <u>58.35</u> / 64.42                       | 53.93 / 55.85                                     |
| Entropy gain (deterministic)                | <u>50.17</u> / <u>50.37</u>                       | 51.04 / <u>53.28</u>                              | 58.40 / <b>64.63</b>                       | 53.20 / <u>56.10</u>                              |
| Student entropy (sampled once, $k = 1024$ ) | 41.35 / 40.11                                     | 39.44 / 39.07                                     | 50.17 / 52.87                              | 43.65 / 44.01                                     |
| Student entropy (sampled once, $k = 2048$ ) | 42.19 / 41.06                                     | 43.18 / 43.02                                     | 53.20 / 52.62                              | 46.19 / 45.57                                     |
| Student entropy (sampled once, $k = 3072$ ) | 42.02 / 42.81                                     | <b>56.77</b> / 40.86                              | 56.77 / 55.90                              | 51.86 / 46.52                                     |
| Entropy gain (sampled, ours)                | <b>52.58</b> $\pm$ 1.94 / <b>52.08</b> $\pm$ 2.28 | <u>56.64</u> $\pm$ 3.79 / <b>56.86</b> $\pm$ 2.32 | <b>62.22</b> $\pm$ 2.42 / 64.07 $\pm$ 4.52 | <b>57.15</b> $\pm$ 1.01 / <b>57.67</b> $\pm$ 1.19 |

Table 1: Distillation pipeline last epoch accuracy (4096-token / 2,048-token thinking budget) on MMLU-Pro.

**Hardware:** We run all experiments<sup>4</sup> on a single A100/H100<sup>5</sup> GPU with 80GB of VRAM to validate the method in resource-constrained environments.

#### 4.1.1 Our method and results

We evaluate the proposed pipeline for 200 epochs (205k questions seen) and 2 thinking budgets (2048 and 4096 tokens). It is compared at similar data budget with:

*Direct reasoning trace, Corrected answer* - naive distillation on the most promising types of reasoning traces (see 4.3);

*Student entropy (deterministic), Entropy gain (deterministic)* - alternative pipeline with weighted sampling replaced with greedy sampling;

*Student entropy (sampled once,  $k=x$ )* - distillation executed against  $k$  samples selected once by student entropy (not recursively at each epoch).

*Entropy gain (sampled), Student entropy (deterministic), Entropy gain (deterministic)* use the proposed data mix (reasoning traces/single-token answers = 1024/256), other methods use only reasoning traces.

Notably, the proposed data mix increases the number of training tokens only by 0.003% - 256 extra tokens compared to 8,388,608 tokens in 1024 reasoning chains.

All randomly sampled methods use 3 random

seeds, reporting  $\pm 95\%$  confidence-intervals, and deterministic methods are evaluated once with greedy sampling.

#### 4.2 Main Results

Table 1 and Figure 2 show the results for MMLU Pro. We see that the proposed method produces a significant improvement over naive distillation (up to 10% gain). For both reasoning budgets, it provides the best result on average across three models in comparison with other approaches. Greedy sampling demonstrates subpar and less stable curves, though surpassing the baselines. Greedy distillation based on student entropy sampled once (Goncharov et al., 2026) shows signs of severe degradation after the initial uplift. However, Qwen 2.5 3B, notably, demonstrates the best performance with it at  $k=3072$ , but degrades at smaller sample sizes, revealing the fragility of the approach.

We also note that our method helps with the model calibration significantly enhancing ROC AUC of single-token entropy for the first half of training. Shortly after, it starts declining which is loosely correlated with the method’s plateau.

#### 4.3 Synthetic Traces Generation

In resource-constrained environments, available VRAM imposes a strict limitation on the length of distilled reasoning chains. As soon we start truncating them, we face a new challenge - which part

<sup>4</sup>Training hyperparameters can be found in Appendix A.

<sup>5</sup>We had to switch cards depending on availability

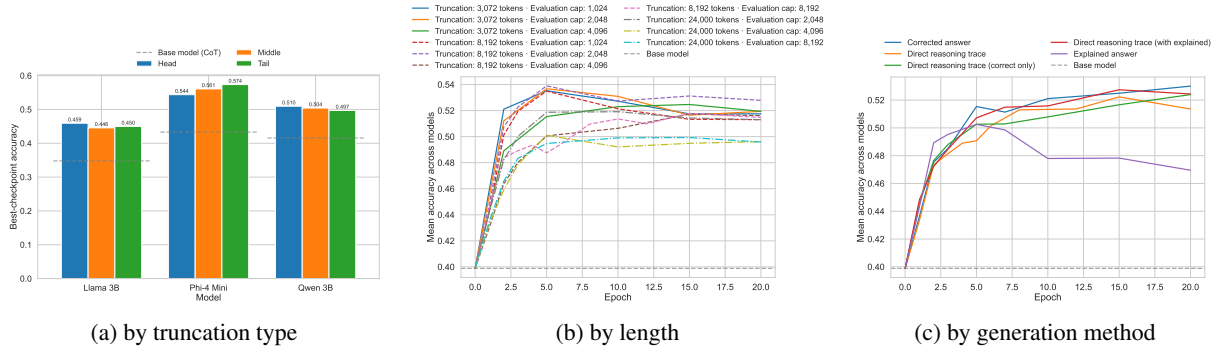


Figure 3: Comparison of synthetic traces across three models (average)

of the trace contains the most useful signal? At last, how exactly do we prompt the teacher model to generate reasoning traces?

We evaluate reasoning traces by: (a) *truncation type* - which part of the trace we keep when it exceeds the maximum length; (b) *length* - what is the optimal training truncation limit and evaluation thinking budget (in tokens). (c) *generation method* (below).

Generation methods (details in Appendix C): (1) *Direct reasoning trace* - baseline prompt to just answer the question; (2) *Direct reasoning trace (correct only)* - same as (1), but we keep only traces with correct answers; (3) *Explained answer* - we give a correct answer to the model and prompt it to explain it; (4) *Direct reasoning trace (with explained)* - same as (1), but replace incorrect traces with (3); (5) *Corrected answer* - same as (1), but for incorrect traces we give the model correct answers and ask it to update its reasoning.

Figure 3 shows that models have a slight preference towards head truncation, medium to short reasoning chains in training and thinking budgets. Surprisingly, longer reasoning chains and greater thinking budgets lead to notable performance decrease. As to the generation methods, we see that augmented reasoning traces lead to elevated performance, while the consistent data leakage results in degradation.

#### 4.4 Sensitivity Study

To attribute performance gains to either the data mix or the entropy-based score and sampler, we compare the proposed pipeline (*entropy gain*) with: (1) alternative pipelines where we sample according to *student entropy* and *discounted entropy gain* (see below); (2) alternative pipeline where we use a random sampler.

*Discounted entropy gain* is defined as

| Sampler                           | Three-model mean                   |
|-----------------------------------|------------------------------------|
| Direct reasoning trace (baseline) | 51.61                              |
| Random                            | $55.08 \pm 0.80$                   |
| Student entropy                   | $55.87 \pm 1.40$                   |
| Discounted entropy gain           | <u>55.98</u>                       |
| Entropy gain                      | <b><math>57.15 \pm 1.01</math></b> |

Table 2: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)

$\max\{H_S(x) - k(t) * H_P(x), 0\}, k(t) \in [0, 1]$ , where  $t$  - training epoch. We set  $k(t)$  initially to 0.8 and decrease it according to a schedule, hypothesizing that as the student undergoes training the static medium model stops being a good aleatoric proxy.

Table 2 shows that across three models on average entropy gain provides the best results confirming the contribution of the aleatoric uncertainty approximation. At the same time, we notice that random sampler performs exceptionally well (confirming findings of Okanovic et al. (2023)) compared to the baseline, which points to the proposed data mix as a significant contributor to the gains.

## 5 Conclusion and Discussion

This paper introduces a fully automated recursive offline distillation method rooted in epistemic uncertainty approximation. It achieves  $57.15 \pm 1.01$  mean accuracy across three students on MMLU-Pro vs 51.61 for direct distillation at a comparable training data budget and offers a much steeper learning curve (baseline score at 15% of its data budget). We confirm that both the entropy difference sampler and the data mix contribute to the gains. Surprisingly, shorter traces and reasoning budgets exhibit better performance. Further exploration of the more rigorous epistemic uncertainty estimation and the surprising effect of such a small data mix augmentation remain future work.

## Limitations

- Method is tested only with small models. We do not know how well it is going to generalize to larger LLMs. Additionally, it might be challenging to find a suitable aleatoric proxy for larger students.
- We use a single dataset. We do not know how well it translates to other settings.
- Alternative aleatoric approximations, such as agreement of the ensemble and other, are left unexplored.
- We did not test and compare multiple sizes of aleatoric proxies and their impact on performance.
- Different compositions of the data mix are not tested due to lack of resources.
- We tested only a single schedule for *Discounted Entropy Gain*. Alternatives are yet to explore.
- Some low-resource environments might have even more constraints and therefore struggle to fit a suitable aleatoric proxy into VRAM.

## References

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. 2024. [On-policy distillation of language models: Learning from self-generated mistakes](#). In *International Conference on Learning Representations*, volume 2024, pages 21246–21263.
- Shengnan An, Zexiong Ma, Zeqi Lin, Nanning Zheng, Jian-Guang Lou, and Weizhu Chen. 2024. [Learning from mistakes makes llm better reasoner](#). *Preprint*, arXiv:2310.20689.
- Lihu Chen and Gaël Varoquaux. 2026. [What is the role of small models in the llm era: A survey](#). *Preprint*, arXiv:2409.06857.
- Daixuan Cheng, Shaohan Huang, and Furu Wei. 2024. [Adapting large language models via reading comprehension](#). In *International Conference on Learning Representations*, volume 2024, pages 48624–48652.
- DeepSeek-AI, Anyi Xu, Bangcai Lin, Bing Xue, Bingxuan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang, Chaofan Lin, Chen Dong, Chenchen Ling, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chengyu Hou, Chenhao Xu, Chenze Shao, Chong Ruan, Conner Sun, and 300 others. 2026. [Deepseek-v4: Towards highly efficient million-token context intelligence](#). *Preprint*, arXiv:2606.19348.
- Stefan Depeweg, Jose-Miguel Hernandez-Lobato, Finale Doshi-Velez, and Steffen Udluft. 2018. [Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning](#). In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1184–1193. PMLR.
- Andrey Goncharov, Daniil Vyazhev, Petr Sychev, Edvard Khalafyan, and Alexey Zaytsev. 2026. [Complexity-aware fine-tuning](#). *Preprint*, arXiv:2506.21220.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2024. [Minillm: Knowledge distillation of large language models](#). In *International Conference on Learning Representations*, volume 2024, pages 32694–32717.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. [Distilling the knowledge in a neural network](#). *Preprint*, arXiv:1503.02531.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. [Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, Toronto, Canada. Association for Computational Linguistics.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. [Lora: Low-rank adaptation of large language models](#). *Preprint*, arXiv:2106.09685.
- Yaning Jia, Chunhui Zhang, Xingjian Diao, Xiangchi Yuan, Zhongyu Ouyang, Chiyu Ma, and Soroush Vosoughi. 2026. [What makes a good curriculum? disentangling the effects of data ordering on LLM mathematical reasoning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 34472–34488, San Diego, California, United States. Association for Computational Linguistics.
- Wangyi Jiang, Yaojie Lu, Hongyu Lin, Xianpei Han, and Le Sun. 2025. [Teach small models to reason by curriculum distillation](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7412–7422, Suzhou, China. Association for Computational Linguistics.
- Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. 2020.

TinyBERT: Distilling BERT for natural language understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4163–4174, Online. Association for Computational Linguistics.

Hoang Anh Just, Myeongseob Ko, and Ruoxi Jia. 2026. The signal is in the steps: Local scoring for reasoning data selection. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 55125–55155. PMLR.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *Preprint*, arXiv:2001.08361.

Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. 2024. DistiLLM: Towards streamlined distillation for large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 24872–24895. PMLR.

Jierui Li and Raymond Mooney. 2024. Distilling algorithmic reasoning from LLMs via explaining solution programs. *Preprint*, arXiv:2404.08148.

Liunian Harold Li, Jack Hessel, Youngjae Yu, Xiang Ren, Kai-Wei Chang, and Yejin Choi. 2023. Symbolic chain-of-thought distillation: Small models can also “think” step-by-step. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2665–2679, Toronto, Canada. Association for Computational Linguistics.

Yang Li, Youssef Emad, Karthik Padthe, Jack Lanchantin, Weizhe Yuan, Thao Nguyen, Jason Weston, Shang-Wen Li, Dong Wang, Ilia Kulikov, and Xian Li. 2025. Naturalthoughts: Selecting and distilling reasoning traces for general reasoning tasks. *Preprint*, arXiv:2507.01921.

Wei Lu, Rachel K. Luu, and Markus J. Buehler. 2024. Fine-tuning large language models for domain adaptation: Exploration of training strategies, scaling, model merging and synergistic capabilities. *Preprint*, arXiv:2409.03444.

Andrey Malinin and Mark Gales. 2021. Uncertainty estimation in autoregressive structured prediction. *Preprint*, arXiv:2002.07650.

Microsoft, :, Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, Dong Chen, Dongdong Chen, Junkun Chen, Weizhu Chen, Yen-Chun Chen, Yi ling Chen, Qi Dai, and 57 others. 2025. Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. *Preprint*, arXiv:2503.01743.

Patrik Okanovic, Roger Waleffe, Vasilis Mageirakos, Konstantinos E. Nikolakakis, Amin Karbasi, Dionysis Kalogieras, Nezihe Merve Gürel, and Theodoros Rekatsinas. 2023. Repeated random sampling for minimizing the time-to-accuracy of learning. *Preprint*, arXiv:2305.18424.

Qwen-Team. 2024. Qwen2.5: A party of foundation models.

Bakbergen Ryskulov, Iker García-Ferrero, David Montero, David Jansen, Ali Hashemi, Jezabel R. Garcia, Antonio Tiene, and Román Orús. 2026. Efficient knowledge distillation for llms: Offline top-k logits and a fused chunked kl loss. *Preprint*, arXiv:2608.03796.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2020. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *Preprint*, arXiv:1910.01108.

Zhongwei Wan, Xin Wang, Che Liu, Samiul Alam, Yu Zheng, Jiachen Liu, Zhongnan Qu, Shen Yan, Yi Zhu, Quanlu Zhang, Mosharaf Chowdhury, and Mi Zhang. 2024. Efficient large language models: A survey. *Preprint*, arXiv:2312.03863.

Haozhe Wang, Haoran Que, Qixin Xu, Minghao Liu, Wangchunshu Zhou, Jiazhan Feng, Wanjun Zhong, Wei Ye, Tong Yang, Wenhao Huang, Ge Zhang, and Fangzhen Lin. 2025a. Reverse-engineered reasoning for open-ended generation. *Preprint*, arXiv:2509.06160.

Tianyang Wang, Yunze Wang, Jun Zhou, Benji Peng, Xinyuan Song, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Ming Liu, Yichao Zhang, Cheng Fei, Caitlyn Heqi Yin, and Lawrence KQ Yan. 2025b. From aleatoric to epistemic: Exploring uncertainty quantification techniques in artificial intelligence. *Preprint*, arXiv:2501.03282.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Preprint*, arXiv:2406.01574.

Shuyao Xu, Cheng Peng, Jiangxuan Long, Weidi Xu, Wei Chu, and Yuan Qi. 2026. Harnessing negative signals: Reinforcement distillation from teacher data for LLM reasoning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1618–1639, San Diego, California, United States. Association for Computational Linguistics.

Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. 2024. A survey on knowledge distillation of large language models. *Preprint*, arXiv:2402.13116.

Cehao Yang, Xueyuan Lin, Xiaojun Wu, Chengjin Xu, Xuhui Jiang, Honghao Liu, Hui Xiong, and Jian Guo. 2026. [Select2Reason: Efficient instruction-tuning data selection for long-CoT reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6656–6671, San Diego, California, United States. Association for Computational Linguistics.

Ling Yang, Zhaochen Yu, Tianjun Zhang, Minkai Xu, Joseph E Gonzalez, Bin CUI, and Shuicheng YAN. 2025. [Supercorrect: Advancing small llm reasoning with thought template distillation and self-correction](#). In *International Conference on Learning Representations*, volume 2025, pages 3386–3418.

Mengxiang Zhang and Lingyuan Liu. 2026. [Calibrated progressive distillation: Co-designing curriculum and target mixing for knowledge distillation of large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6757–6776, San Diego, California, United States. Association for Computational Linguistics.

Yifei Zhang, Bo Pan, Chen Ling, Yuntong Hu, and Liang Zhao. 2024. [ELAD: Explanation-guided large language models active distillation](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4463–4475, Bangkok, Thailand. Association for Computational Linguistics.

Qinhong Zhou, Zonghan Yang, Peng Li, and Yang Liu. 2023. [Bridging the gap between decision and logits in decision-based knowledge distillation for pre-trained language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13234–13248, Toronto, Canada. Association for Computational Linguistics.

## A Experimental Hyperparameters

### A.1 Optimization and Batching

| Hyperparameter              | Value               |
|-----------------------------|---------------------|
| Optimizer                   | AdamW (adamw_torch) |
| Learning rate               | $10^{-4}$           |
| Learning-rate schedule      | Cosine              |
| Warmup ratio                | 0.06                |
| Weight decay                | 0                   |
| Adam ( $\beta_1, \beta_2$ ) | (0.9, 0.999)        |
| Adam epsilon                | $10^{-8}$           |
| Maximum gradient norm       | 1.0                 |
| Mixed precision             | BF16 enabled        |
| Gradient checkpointing      | Enabled             |
| torch.compile               | Enabled             |

### A.2 LoRA and Trainable Parameters

| Hyperparameter          | Value    |
|-------------------------|----------|
| LoRA rank $r$           | 8        |
| LoRA alpha              | 16       |
| LoRA scaling $\alpha/r$ | 2        |
| LoRA dropout            | 0.1      |
| Initialization          | Gaussian |
| Bias adaptation         | None     |

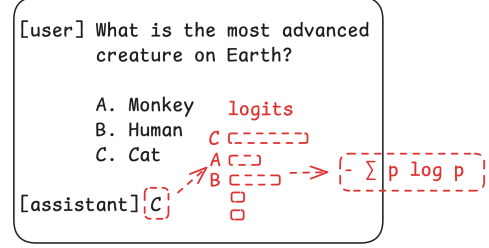


Figure 4: Single-token entropy example

## B Single-token entropy

Single-token entropy (Figure 4) measures a model’s uncertainty over the first answer token when prompted to answer a question with a single token. For a question  $x$ , we apply softmax to the first-token logits to obtain probabilities  $p(v | x)$  over the model’s full vocabulary  $V$ , and compute normalized Shannon entropy:

$$H(x) = -\frac{1}{\log |V|} \sum_{v \in V} p(v | x) \log p(v | x). \quad (2)$$

Dividing by  $\log |V|$  puts student and proxy entropies on a common  $[0, 1]$  scale despite differences in vocabulary size: values near zero indicate a concentrated prediction, while values near one indicate a nearly uniform distribution.

## C Synthetic trace generation methods

| Sampler                           | LLaMA-3B                  | Qwen-3B                   | Phi4-mini                 | Mean                      |
|-----------------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Direct reasoning trace (baseline) | 46.09                     | 51.45                     | 57.27                     | 51.61                     |
| Random                            | $51.77 \pm 2.76$          | $\mathbf{57.91} \pm 1.40$ | $55.57 \pm 2.54$          | $55.08 \pm 0.80$          |
| Student entropy                   | $51.18 \pm 3.11$          | $57.32 \pm 1.52$          | $59.12 \pm 5.22$          | $55.87 \pm 1.40$          |
| Discounted entropy gain           | 51.50                     | 56.90                     | $59.53 \pm 2.11$          | 55.98                     |
| Entropy gain                      | $\mathbf{52.58} \pm 1.94$ | $56.64 \pm 3.79$          | $\mathbf{62.22} \pm 2.42$ | $\mathbf{57.15} \pm 1.01$ |

Table 3: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)